Domain decomposition

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$$\begin{cases} -\Delta u_1^k = f & \text{sur } \Omega_1 \\ u_1^k = g & \text{sur } \partial\Omega_1 \setminus (\{\beta\} \times (0,1)) \\ u_1^k(\beta, y) = u_2^{k-1}(\beta, y) \end{cases} \tag{1}$$

$$\begin{cases} -\Delta u_2^k = f & \text{sur } \Omega_2 \\ u_2^k = g & \text{sur } \partial\Omega_2 \setminus (\{\alpha\} \times (0,1)) \\ u_2^k(\alpha, y) = u_1^k(\alpha, y) \end{cases} \tag{2}$$

On pose

$$e_i^k = u - u_i^k$$

$$\begin{cases} -\Delta e_1^k = 0 & \text{sur } \Omega_1 \\ e_1^k = 0 & \text{sur } \partial\Omega_1 \setminus (\{\beta\} \times (0,1)) \\ e_1^k(\beta, y) = e_2^{k-1}(\beta, y) \end{cases} \tag{3}$$

$$\begin{cases} -\Delta e_2^k = 0 & \text{sur } \Omega_2 \\ e_2^k = 0 & \text{sur } \partial\Omega_2 \setminus (\{\alpha\} \times (0,1)) \\ e_2^k(\alpha, y) = e_1^k(\alpha, y) \end{cases} \tag{4}$$

On pose pour tout $n \in \mathbb{N}$:

$$z_n(y) = \sqrt{2} \sin(n\pi y)$$

La famille $(z_n)_{n\in\mathbb{N}}$ forme une base hilbertienne de $L^2(0,1)$.

$$e_i^k(x,y) = \sum_{n=1}^{\infty} e_{i,n}^k(x)\, z_n(y)$$

En appliquant (3), on obtient :

$$\sum_{n=1}^{\infty} \Big(\partial_{xx} e_1^k(x) - n^2\pi^2 e_{1,n}^k z_n(y) \Big) = 0$$

Donc

$$\partial_{xx} e^k_{1,n} - n^2\pi^2 e^k_{1,n} = 0$$

Donc

$$e^k_{1,n}(x) = A_n e^{n\pi x} + B_n e^{-n\pi x}$$

$$\begin{aligned}
\text{or} \quad & e_1(0,y) = 0 \\
\text{donc} \quad & \sum_{n=1}^{\infty} (A_n + B_n)\sin(n\pi y) = 0 \\
\text{donc} \quad & A_n = -B_n \\
\text{en } x = \beta : \quad & \sum_{n=1}^{\infty} A_n\big(e^{n\pi\beta} - e^{-n\pi\beta}\big)\sin(n\pi y) = e_2^{k-1}(\beta, y)
\end{aligned}$$

Donc

$$A_n\big(e^{n\pi\beta} - e^{-n\pi\beta}\big) = e^{k-1}_{2,n}(\beta)$$

Donc

$$A_n = \frac{e^{k-1}_{2,n}(\beta)}{e^{n\pi\beta} - e^{-n\pi\beta}}$$

De plus, en appliquant le même raisonnement à e_2^k, on obtient :

$$e^k_{2,n} = C_n e^{n\pi x} + D_n e^{-n\pi x}$$

or

$$e_2^k(b,y) = 0 \quad \forall y \in (0,1) \quad \text{donc} \quad D_n = -C_n e^{2n\pi b}$$

et $e_2^k(\alpha, y) = e_1^k(\alpha, y)$ donc

$$\begin{aligned}
&\sum_{n=1}^{\infty} C_n\big(e^{n\pi\alpha} - e^{n\pi(2b-\alpha)}\big)\sin(n\pi y) \\
&= \sum_{n=1}^{\infty} A_n\big(e^{n\pi\alpha} - e^{-n\pi\alpha}\big)\sin(n\pi y)
\end{aligned}$$

Donc

$$C_n = \frac{e^{k-1}_{2,n}(\beta)}{e^{n\pi\beta} - e^{-n\pi\beta}} \cdot \frac{e^{n\pi\alpha} - e^{-n\pi\alpha}}{e^{n\pi\alpha} - e^{n\pi(2b-\alpha)}}$$

Enfin

$$e_2^k(\beta, y) = \sum_{n=1}^{\infty} e^{k-1}_{2,n}(\beta)\frac{e^{n\pi\alpha} - e^{-n\pi\alpha}}{e^{n\pi\beta} - e^{-n\pi\beta}}\,\frac{e^{n\pi\beta} - e^{n\pi(2b-\beta)}}{e^{n\pi\alpha} - e^{n\pi(2b-\alpha)}}$$